Research Proposal: Precision Manipulation in Robotic Upper Limbs through Multi-Modal Sensing

Zilin Chen, PHD Proposal in Dept. of ECE, HKUST

1 Introduction

The development of dexterous robotic manipulation remains a fundamental challenge in robotics research. While significant progress has been made in visual perception for grasping, the integration of tactile and force feedback for precision manipulation of novel objects remains underdeveloped. This research proposal aims to address the gap in robotic manipulation by developing novel control strategies that leverage multi-modal force sensing to enable precise manipulation of various unknown objects in unstructured environments.
Human manipulation capabilities far exceed current robotic systems, particularly in handling objects with varying material properties, shapes, and weights. This research will explore how advanced force sensing and control algorithms can bridge this gap, enabling robots to perform delicate manipulation tasks with human-like proficiency.

2 Literature Review

Nowadays, walking is nearly a basic skill for all humanoid robots. Researchers and corporate researchers use all kinds of mechine learing methods to teach the robots how to walk, just like Ilija Radosavovic did in his research [Radosavovic et al., 2024]. Using reinforcement learing can easily teach a humanoid-robot to walk on several basic terrain. However, the robots' upper limbs, especially how we control the robots' hands to achieve human-level dexterity is still an ongoing challenge [Tong et al., 2024].

3 Research Objectives

The overarching goal of my research is to advance the dexterity, adaptability, and intelligence of robotic hands by tightly integrating high-fidelity sensing, feedback-driven control, and data-driven learning. Specifically, the objectives are:

1. Design and implementation of an advanced sensing system — with an emphasis on high-resolution, multi-axis force sensing embedded in robotic hands. The aim is to capture rich, real-time interaction data during dynamic manipulation tasks, enabling robots to perceive subtle contact forces, slip, and object compliance.

2. Development of sensor-fusion control algorithms, integrating force feedback with visual perception to achieve closed-loop, adaptive grasping and manipulation. These algorithms will focus on enhancing grasp stability, accuracy, and sensitivity, allowing robotic hands to respond intelligently to unexpected disturbances or variations in object geometry.

3. Creation of learning-based frameworks for generalizable manipulation, leveraging machine learning methods to transfer manipulation skills across diverse object categories, shapes, and material properties. Unlike traditional approaches constrained to pre-marked or well-modeled objects, the proposed methods will enable robots to handle unknown items with minimal prior information.

4. Comprehensive experimental validation on physical robotic platforms, conducting rigorous real-world tests to verify the robustness, efficiency, and scalability of the proposed sensing, control, and learning strategies, ensuring that theoretical advances translate into tangible performance gains.

4 Methodology

The proposed research will follow a systematic approach that combines modeling, simulation, learning-based policy development, and real-world system integration. The methodology is structured into three major phases:

4.1 Modeling and Simulation Studies

The research will begin with the development of accurate kinematic and dynamic models of the robotic hand and arm. Force-sensing data will serve as the primary input to analyze contact interactions during grasping. Physics-based simulations will be conducted to study grasp stability, contact force distribution, and object discrimination. These studies in the simulation environment will guide the design of control strategies and help identify the critical sensing modalities required for high-precision manipulation.

4.2 Learning-Based Grasp Policy Development

To achieve robust and generalizable manipulation capabilities, learning-based methods will be employed in parallel with the modeling efforts. Reinforcement learning will be used to train grasping policies in a simulated environment, enabling the robot to handle diverse objects with varying sizes, shapes, and material properties. Domain adaptation techniques will be explored to transfer the learned policies to the physical robotic system, while online learning will allow the robot to refine its performance continuously during real-world tasks.

4.3 System Integration and Experimental Validation

Finally, the developed sensing, control, and learning components will be integrated into a complete robotic manipulation platform. The multi-fingered hand, robotic arm, and multimodal perception system will be combined to form a unified testbed. Comprehensive experiments will be performed to validate the proposed methods, benchmarking grasp success rates, force modulation accuracy, and adaptability against state-of-the-art robotic manipulation frameworks.

5 Expected Contributions

This research is expected to generate several key contributions to the field of robotic manipulation and humanoid robotics:

- **A novel force-based manipulation control algorithm:** A new control strategy will be developed to leverage high-resolution tactile feedback, substantially enhancing the accuracy, stability, and consistency of humanoid robot upper-limb grasping and manipulation.
- **An integrated multimodal sensing framework:** A unified platform will be established to fuse tactile, force, and visual data, providing rich and reliable information to support the development of next-generation humanoid robot systems.
- **Advancing real-world deployment of humanoid robots:** The proposed methods will improve the adaptability and robustness of humanoid robots in unstructured environments, enabling practical applications that go beyond proof-of-concept demonstrations.

6 Timeline

- **Year 1**: Conduct an in-depth literature review to identify current challenges in tactile sensing, force control, and learning-based manipulation. Establish a robotic hardware platform by integrating the multi-fingered hand, robotic arm, and vision system. Implement the data acquisition and processing infrastructure. Perform preliminary experiments to verify sensing accuracy, system calibration, and baseline performance.

- **Year 2**: Develop the core control and learning algorithms, including hybrid control strategies and reinforcement learning policies. Perform extensive simulation studies to validate stability, safety, and adaptability. Collecting enough experimental data in both simulation and physical environment. Begin domain adaptation research to prepare for transferring learned policies from simulation to physical hardware.
- **Year 3**: Deploy the developed algorithms onto the real robotic platform. Conduct extensive experimental validation across a wide range of manipulation tasks and object types. Benchmark performance against state-of-the-art methods using quantitative metrics such as grasp success rate, force modulation accuracy, and real-time responsiveness.
- **Year 4**: Refine the integrated system based on experimental findings. Finalize the dissertation with comprehensive documentation of methodologies, results, and theoretical insights. Complete thesis defense and disseminate research outcomes to the broader robotics community.

7 Conclusion

This research proposal outlines a comprehensive approach to advancing robotic manipulation capabilities through innovative use of force sensing technology. By combining rigorous theoretical development with practical implementation, this work aims to significantly advance the state of the art in robotic manipulation and contribute to the broader field of embodied intelligence. The proposed research aligns perfectly with Professor Shen's focus on embodied intelligence and precision upper limb control, offering exciting opportunities for fundamental advances with practical applications.

References

[Radosavovic et al., 2024] Radosavovic, I., Xiao, T., Zhang, B., Darrell, T., Malik, J., and Sreenath, K. (2024). Real-world humanoid locomotion with reinforcement learning. *Science Robotics*, 9(89):eadi9579.

[Tong et al., 2024] Tong, Y., Liu, H., and Zhang, Z. (2024). Advancements in humanoid robots: A comprehensive review and future prospects. *IEEE/CAA Journal of Automatica Sinica*, 11(2):301–328.